

Ruoyan (Tristan) Feng

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EDUCATION

Cornell University

Master of Engineering in Financial Engineering, GPA: 4.15

Ithaca, NY

Dec. 2025

- **Relevant Courses:** Machine Learning, Stochastic Calculus, Derivatives Securities, Fixed Income, Optimization Models, Risk Management, Algorithm Trading, Monte Carlo Simulation, Energy

New York University

B.A. in Mathematics and Economics, Minor in Computer Science and Business, GPA: 3.77

New York, NY

May 2024

- **Relevant Courses:** Data Management Analysis, Computer System Organization, Time Series, Theory of Probability, Math of Finance, Differential Equation, Numerical Analysis, Real Analysis

SKILLS

Technical: Python, SQL, Linux, Tableau, VBA, R, MATLAB, Microsoft Office, Bloomberg Terminal

PROFESSIONAL EXPERIENCE

Quantitative Research Intern, Zhurun Investment Co., Ltd., Shanghai, China

June – Aug. 2024

- Used CVaR as a factor for stock selection in CSI300 to enhance portfolio risk-adjusted return during left tail events; leveraged tick level data, VWAR computation, and parallelized data cleaning; back-tested it via stress scenario analysis
- Engineered predictive cross-sectional features using Genetic Programming to derive interpretable, robust alpha signals; evolved base factors such as volatility and momentum through tree-based symbolic modeling and fitness filtering
- Maintained a GPT-based LLM model to extract actionable investment insights from financial and earnings reports; fine-tuned on domain-specific text and extracted sentiment, tone, and metrics to feed key signals into the trading model
- Collaborated with senior quants to develop machine learning based trading pipelines, including a LightGBM model with 200+ fundamental and market indicator features; optimized efficiency by 50% utilizing Joblib and Multiprocessing

Quantitative Research CTA Intern, Evolution Labs, Shanghai, China

June – Aug. 2023

- Implemented a futures strategy using Bollinger Bands to capture mispricing and exploit price discrepancies between contract maturities of the same commodity; identified overbought/oversold signals and cross-period arbitrage trades
- Conducted factor research to develop frequency-based trading strategies by applying Fourier Transform in Python to identify dominant frequencies in energy distribution over time; aligned portfolio positions to evolving market dynamics
- Designed factor evaluation functions, validated the trading strategy through a 5-year backtesting process, demonstrating improved responsiveness to intraday trends and achieving a Sharpe ratio of 1.27 and an average annual return of 24.2%

Financial Representative Assistant, Northwestern Mutual, New York, NY

June – Aug. 2022

- Analyzed trends in Equities, Fixed Income, and Real Estate to test inflation hedge ability under interest rate stress scenarios; built regression models to optimize portfolio allocation and enhance resilience to macroeconomic shocks
- Included least correlated equity assets in a portfolio and applied Monte Carlo Simulation to estimate long-term risk exposure; identified them by computing serial correlation in Python using historical return to capture idiosyncratic risk
- Mimicked the Black-Scholes model with factors including age, job and financial impact to price keyman insurance using VBA; utilized sensitivity analysis on the pricing model to compute “Greeks” of the insurance premium

PROJECTS

Algorithmic Execution Modeling Project, Ithaca, NY

Mar. – May 2025

- Built a weighted algorithmic trading model by integrating momentum, hidden order, GARCH, and imbalance signals derived from tick-level data; derived real-time features to capture short-term price dynamics and market microstructure
- Back tested execution algorithm using out-of-sample trade simulations; consistently outperformed TWAP by over 30% in spread capture per stock, leveraging real-time signals and adaptive thresholds for improved execution precision

Commodity Signal Analysis Project, New York, NY

Mar. – May 2024

- Analyzed market dynamics by combining real data from S&P 500 and wheat prices from FRED with simulated signals generated via Geometric Brownian Motion for stocks and sinusoidal functions for commodities under stochastic models
- Applied Discrete Fourier Transform(DFT) to evaluate the frequency components of financial signals, demonstrating its ability in capturing cyclical trends in commodity while highlighting its limitations with non-cyclical stock price pattern

LEADERSHIP & OTHERS

Leadership: Led community workshops on financial markets, advising seniors on budgeting, saving, and fraud prevention

Interests: Motorbiking, Reading, Football, Soccer, Fishing, Camping, Photographing, Fitness